

Portfolio Question 2.1.P - Model Selection

Daniel Lawson University of Bristol

Portfolio 2.1.P (v4.0.1)

Portfolio Example Question

- ▶ Portfolio Question 2.1.P: When is a bigger model better?
 - ▶ If we give more data and parameters to a (regression) model, in what sense does it get better?
 - ▶ How is “model goodness” measured accounting for model complexity?
 - ▶ Hints: take a look at the motivation for Akaike Information Criterion
- ▶ Format
 - ▶ Introduction
 - ▶ Non-trivial implication
 - ▶ Discussion

Model Selection Motivation: Residuals

- ▶ The **residual sum of squares** for n predictions of a univariate y :

$$R^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- ▶ The expected value of the prediction error $\mathbb{E}(e^2) = R^2/n$.
- ▶ What happens if **compare two models** M_1 and M_2 , where M_1 is a subset of M_2 ?

Linear Models - Model selection

- ▶ For illustration, consider

$$Y = \mathbf{x}_1 A_1 + \mathbf{e}_1$$

- ▶ and

$$Y = \mathbf{x}_1 A_1 + \mathbf{x}_2 A_2 + \mathbf{e}.$$

- ▶ Unless $\mathbf{x}_2 = 0$ or $\mathbf{x}_2 \equiv \mathbf{x}_1$, then $\mathbf{e}^2$ will be smaller than $\mathbf{e}_1^2$.
 - ▶ This is an example of a more general rule: **larger (nested) models always have better predictions.**
- ▶ So prediction error is OK to use to fit models with the same dimension, but is incomplete for **model selection.**

Out-of-sample prediction accuracy in linear regression

- For general y'_i , we can rearrange the formula for variance to show:

$$\mathbb{E}(e'^2_i) = \mathbb{E} \left[(y'_i - \hat{y}_i)^T (y'_i - \hat{y}_i) \right] \quad (1)$$

$$= \text{Var}[y'_i] + \text{Var}[\hat{y}_i] - 2\text{Cov}[y'_i, \hat{y}_i] + [\mathbb{E}(y'_i) - \mathbb{E}(\hat{y}_i)]^2 \quad (2)$$

- Now denote out-of-sample $\text{Cov}[y'_i, \hat{y}_i] = 0$ whereas within-sample, $\text{Cov}[y_i, \hat{y}_i] \neq 0$.
- Therefore:

$$\mathbb{E}(e'^2_i) = \mathbb{E}(e^2_i) + 2\text{Cov}[y_i, \hat{y}_i]$$

Quantifying Out-of-sample prediction accuracy

- ▶ The mean out-of-sample prediction error can be rewritten (see Appendix) as:

$$\mathbb{E}(e'^2) = n^{-1} \sum_{i=1}^n e_i'^2 = n^{-1} \sum_{i=1}^n e_i^2 + 2n^{-1} \sigma^2 p$$

- ▶ i.e. the total error is a constant times p more than it should be!
- ▶ This motivates minimising **Akaike's Information Criterion** for a general log-likelihood $\mathbb{L}(\hat{\theta})$:

$$AIC = -2\mathbb{L}(\hat{\theta}) + 2\text{Dim}(\theta)$$

Linear model optimism and AIC

- ▶ **Akaike's Information Criterion** (AIC) penalty $2p$
- ▶ Deviance Information Criterion (DIC) penalty $2p_D$ with
 $p_D = \overline{\text{Var}(\mathbb{L}(\hat{\theta}))}/2$
- ▶ Bayesian Information Criterion (BIC) penalty $\log(n)p$
- ▶ Widely applicable information criterion (WAIC) and WBIC
“asymptotically equivalent to cross-validation loss”
- ▶ etc

Discussion

- ▶ Why all these criteria? Can they all be right?
- ▶ What assumptions are they making?
- ▶ How do they relate to cross-validation?